



Caringbah High School

Year 11 2015

Mathematics

Preliminary HSC Course

Semester 2 Exam

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
- Board-approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- In Questions 6 - 12, show relevant mathematical reasoning and/or calculations
- Marks may not be awarded for partial or incomplete answers

Total marks – 85

Section I 5 marks

Attempt Questions 1-5
Mark your answers on the answer sheet provided. You may detach the sheet and write your name on it.

Section II 80 marks

Attempt Questions 6 - 12
Write your answers on the answer sheets provided. Ensure your name or student number is clearly visible.

Name: _____

Class: _____

Marker's Use Only									
Section I	Section II							Total	
Q 1-5	Q6	Q7	Q8	Q9	Q10	Q11	Q12		
/5	/12	/12	/12	/11	/11	/11	/11	/85	%

Question 1 - 5 (1 mark each) Answer on the page provided.

- 1) What is the value of $\sqrt{\frac{4^2 + 18^2}{23^2 - 18^2}}$, correct to 3 significant figures?
 A) 1.28 B) 1.29 C) 1.287 D) 1.288

- 2) Which equation represents a line perpendicular to $3x - 2y = 5$?
 A) $3x - 2y = 10$ B) $3x + 2y = 10$
 C) $2x + 3y = 10$ D) $2x - 3y = 10$

- 3) If $(3 + \sqrt{2})^2 = a + \sqrt{b}$, which is the correct pairing for a and b , given that both a and b are rational?
 A) $a = 13, b = 72$ B) $a = 13, b = 2$
 C) $a = 11, b = 2$ D) $a = 11, b = 72$

- 4) Which of the expressions below is equivalent to $\cos(270^\circ - \theta)$?
 A) $\sin \theta$ B) $-\sin \theta$ C) $\cos \theta$ D) $-\cos \theta$

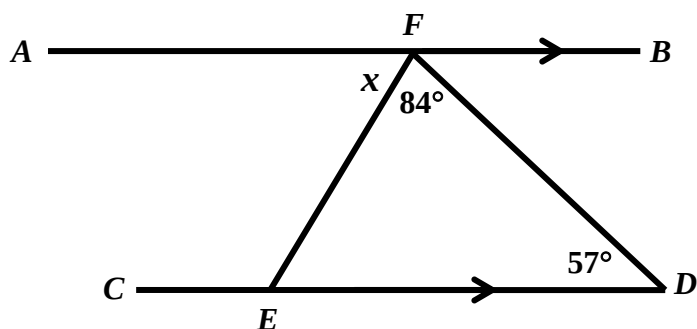
- 5) Which inequality defines the domain of the function $f(x) = \sqrt{1 - x} + \sqrt{x + 4}$?
 A) $x \leq -4$ and $x \geq 1$ B) $x \leq -1$ and $x \geq 4$
 C) $-4 \leq x \leq 1$ D) $-1 \leq x \leq 4$

End of multiple choice questions

Question 6 (12 marks) Start a NEW page.

Marks

- a) Insurance has increased by 10% to \$462. 1
What was the cost of insurance before the increase?
- b) State the range of $y = 1 + 2^x$. 1
- c) Solve $4(3x - 2) - 3(x + 4) = 13$. 2
- d) Factorise fully $2x^2 + 8x + 8$. 2
- e) Find the values of x for which $|3x + 2| \leq 8$. 2
- f) Evaluate $\lim_{x \rightarrow 0} \frac{x^3 - 2x}{x}$. 2
- g) Find the value of x , giving reasons. 2



Question 7 (12 marks) Start a NEW page.

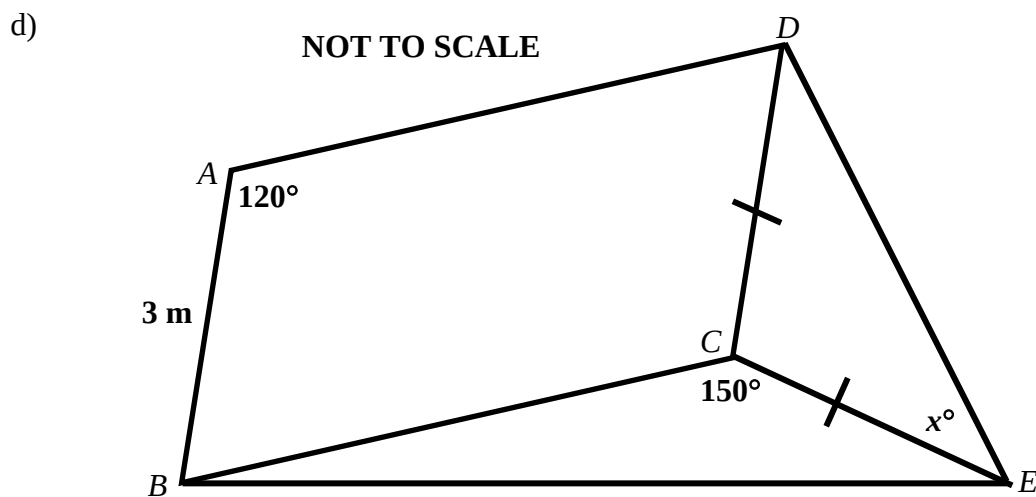
Marks

- a) Solve $x - \frac{x+4}{4} = 2$. 2
- b) Sketch the graph of $y = \sqrt{9-x^2}$ showing all important features. 2
- c) The line $5x + ky = 8$ passes through the point $(-2, 3)$. 2
Find the value of k .
- d) Consider $f(x) = \frac{2}{x+1}$. Find $f\left(\frac{1}{a}\right)$ as a simple fraction. 2
- e) Solve $\sin\theta = -\frac{\sqrt{3}}{2}$ for $0^\circ \leq \theta \leq 360^\circ$. 2
- f) Factorise fully $x^3 - x^2 - x + 1$. 2

Question 8 (12 marks) Start a NEW page.

Marks

- a) Show that $f(x) = x^{2n}$ is an even function. 1
- b) Find the domain of $y = \frac{\sqrt{x+1}}{x-1}$. 2
- c) Shade the region defined by $x^2 + y^2 \leq 4$ 2



$ABCD$ is a parallelogram. $\angle BAD = 120^\circ$ and $\angle BCE = 150^\circ$.

$CD = CE$ and $AB = 3$ metres.

Copy or trace this diagram onto your answer page.

- i) Find the value of x , giving appropriate reasons for each step. 3
- ii) Find the exact length of DE in simplest form. 2
- e) If $g(x) = \frac{1}{x^2} - 9$, find the value(s) of x for which $g(x) = 0$. 2

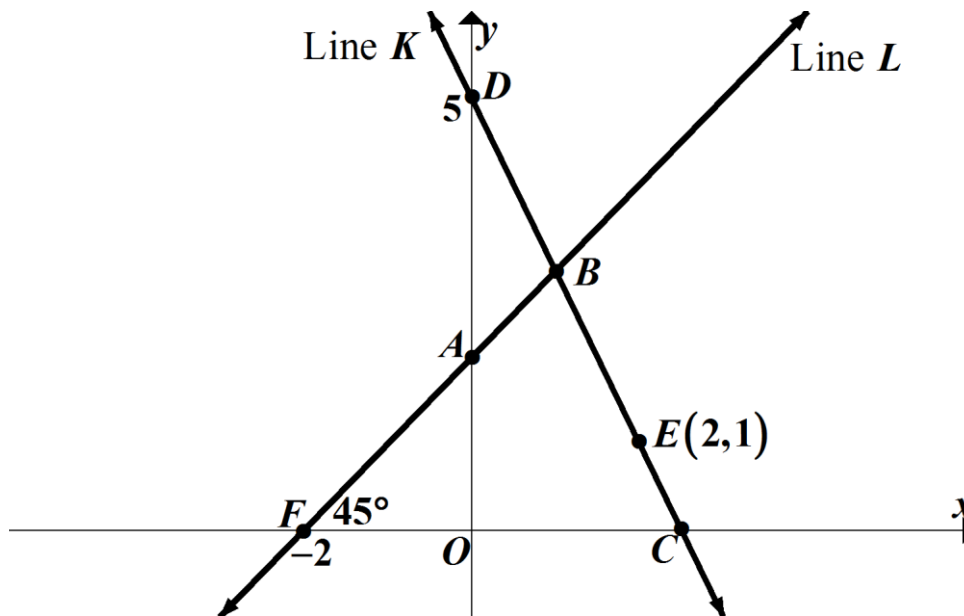
Question 9 (11 marks) Start a NEW page.

Marks

a) Simplify $\frac{(1 - \cos \theta)(1 + \cos \theta)}{\tan^2 \theta}$.

2

b)



The questions below refer to the diagram drawn above.

Copy the diagram onto your answer page.

i) Show that Line K has equation $2x + y - 5 = 0$.

2

ii) Give appropriate reasons to explain why Line L has equation $y = x + 2$.

2

iii) Line K and Line L intersect at the point B.
Find the coordinates of B.

2

iv) Find the coordinates of the point C.

1

v) Hence or otherwise, find the area of quadrilateral OABC.

2

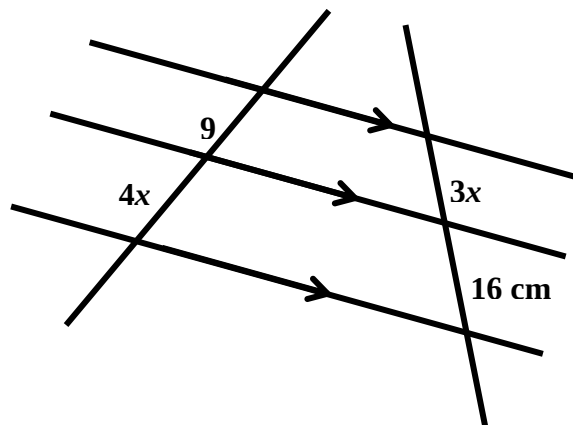
Question 10 (11 marks) Start a NEW page.

Marks

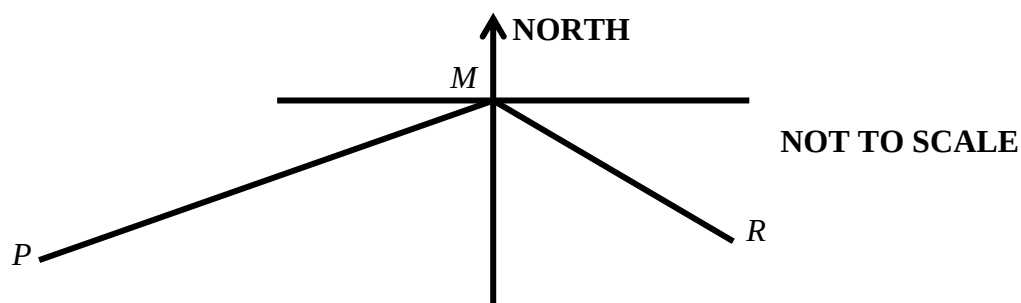
- a) Determine the sum of the interior angles of a regular polygon if each exterior angle is 24° . 2

- b) The line $kx + 3y + 7 = 0$ is perpendicular to the line $y = \frac{5}{2}x + 4$. 2
Find the value of k .

- c) Using the diagram below, find the value of x . (All measurement are in cm) 2



- d) From a town M , town R is 109 km away on a bearing of 110° .
Town P is 220 km away from town M on a bearing of 230° .

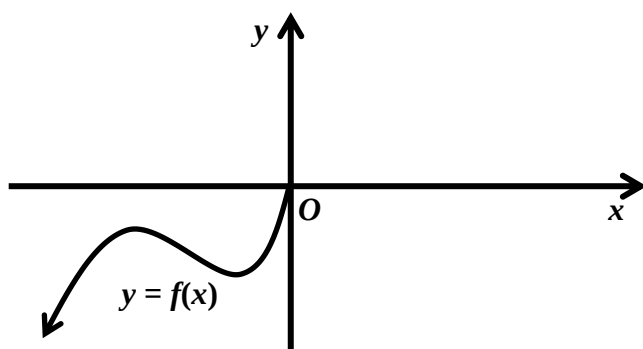


- i) Copy the above diagram onto your answer sheet, add the given information and find the size of $\angle PMR$. 1
- ii) Find the distance between the two towns R and P (nearest km). 2
- e) Sketch $y = \log_2(x - 2)$, showing any important features. 2

Question 11 (11 marks) Start a NEW page.

Marks

- a) $y = f(x)$ has been drawn for $x \leq 0$.



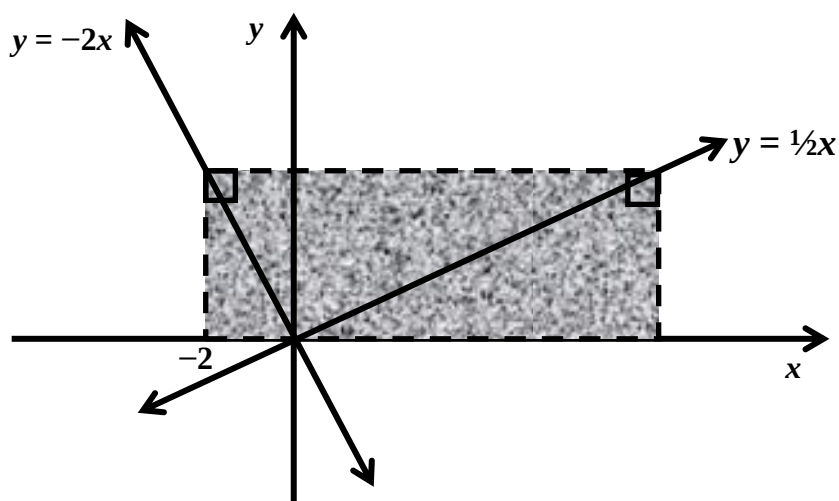
Copy the diagram and complete the graph of $y = f(x)$ if

1

$f(x)$ is an odd function.

- b) Calculate the shaded area.

2



- c) Sketch $y = \frac{x^2 - 1}{x - 1}$ showing any important features.

2

- d) Find the centre and radius of $x^2 + y^2 - 6y - 7 = 0$.

3

- e) If $\tan \theta = \frac{3}{7}$ and $\sin \theta < 0$, find the value of $\cos \theta$ in exact form.

3

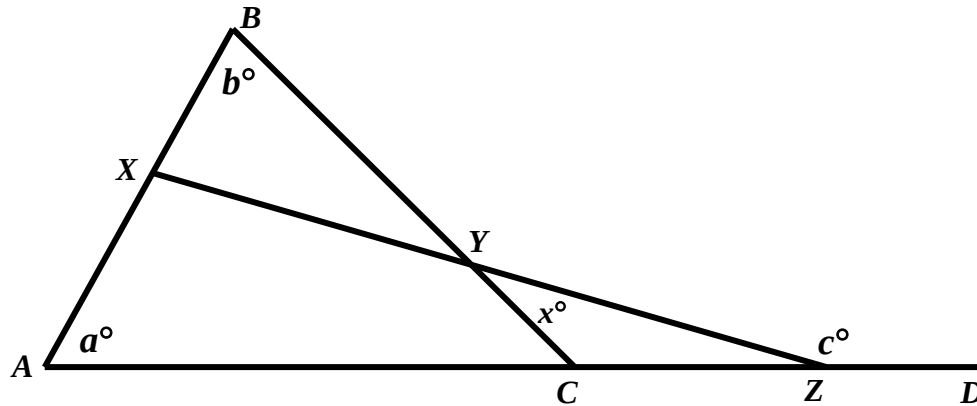
Question 12 (11 marks) Start a NEW page.

Marks

- a) Sketch $y = |2x - 1|$ showing any important features.

2

b)



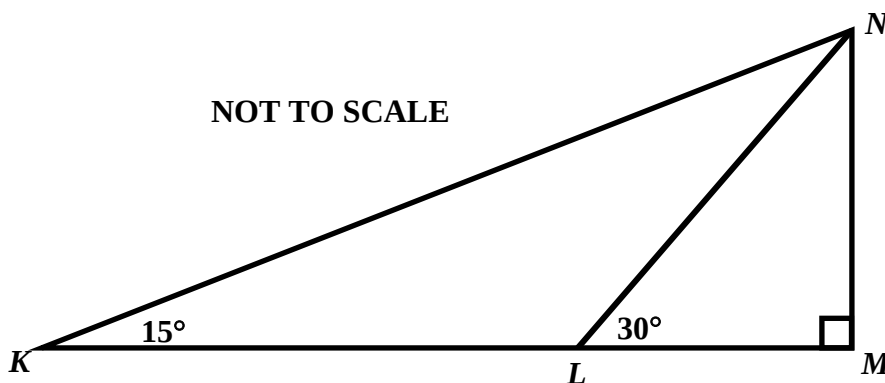
AD , XZ , AB and BC are straight lines. If $\angle CYZ = x^\circ$, find an expression for x in terms of a , b and c .

3

- c) Factorise $16x^3 - 250$.

2

d)



- i) Explain why $KL = LN$.

1

- ii) If $NM = 1$, show that $\tan 15^\circ = 2 - \sqrt{3}$.

3

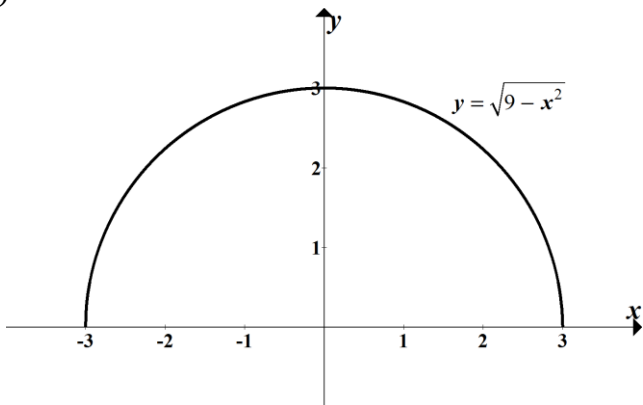
END OF EXAM

Name: _____

Multiple choice answer page. Fill in either A, B, C or D for questions 1-5.

This page must be handed in with your answer booklets

1.	
2.	
3.	
4.	
5.	

CHS YEAR 11 MATHEMATICS (2U) 2015	SEMESTER 2 EXAM SOLUTIONS
Multiple Choice Section: 1.B 2.C 3.D 4.B 5.C Question 1. $1.287841832 = 1.29$ (3 sig fig) -----[B] Question 2. $2x + 3y = 10$ -----[C] Question 3. $(3 + \sqrt{2})^2 = 9 + 6\sqrt{2} + 2$ $= 11 + \sqrt{72}$ -----[D] Question 4. $\cos(270^\circ - \theta) = \cos(180^\circ + (90^\circ - \theta))$ $= -\cos(90^\circ - \theta)$ $= -\sin\theta$ -----[B] Question 5. $\sqrt{1-x} \rightarrow 1-x \geq 0 \rightarrow x \leq 1$ $\sqrt{x+4} \rightarrow x+4 \geq 0 \rightarrow x \geq -4$ $\therefore -4 \leq x \leq 1$ -----[C] Question 6. a) $110\% = \\$462$ $\therefore 100\% = \\$462 \div 1.1 = \\420 b) $R: y > 1$ c) $12x - 8 - 3x - 12 = 13$ $\therefore 9x = 33 \rightarrow x = \frac{11}{3}$	d) $2x^2 + 8x + 8 = 2(x^2 + 4x + 4)$ $= 2(x+2)^2$ e) $-8 \leq 3x + 2 \leq 8$ $-10 \leq 3x \leq 6$ $-\frac{10}{3} \leq x \leq 2$ f) $\lim_{x \rightarrow 0} \frac{x^3 - 2x}{x} = \lim_{x \rightarrow 0} (x^2 - 2)$ $= -2$ g) $\angle FED = 39^\circ$ ($\angle$ sum of $\triangle FED$) $\therefore x = \angle AFE = 39^\circ$ (alt $\angle$'s and $AB \parallel CD$) Question 7. a) $4x - x - 4 = 8$ $3x = 12 \rightarrow x = 4$ b)  c) $\therefore 5 \times -2 + 3k = 8$ by substitution $3k = 18 \rightarrow k = 6$

$$\begin{aligned}
 \text{d) } f\left(\frac{1}{a}\right) &= \frac{2}{\frac{1}{a}+1} \\
 &= \frac{2}{\frac{1+a}{a}} \\
 &= \frac{2a}{1+a}
 \end{aligned}$$

e) Basic angle = 60°

$$\therefore \theta = 180^\circ + 60^\circ \text{ and } \theta = 360^\circ - 60^\circ$$

$$\therefore \theta = 240^\circ \text{ and } \theta = 300^\circ$$

$$\begin{aligned}
 \text{f) } x^2(x-1) - (x-1) &= (x^2-1)(x-1) \\
 &= (x+1)(x-1)(x-1) \\
 &= (x-1)^2(x+1)
 \end{aligned}$$

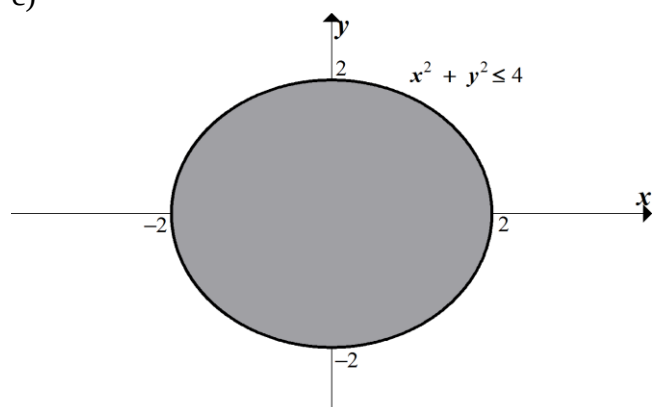
Question 8.

a) For an even function $f(x) = f(-x)$

$$\begin{aligned}
 f(-x) &= \left[(-x)^2\right]^n \\
 &= \left[x^2\right]^n \\
 &= x^{2n} = f(x)
 \end{aligned}$$

b) D: $x \geq -1$, excluding $x = 1$

c)



$$\begin{aligned}
 \text{d) i) } \angle BCD &= 120^\circ \text{ (opp } \angle\text{'s in } \parallel\text{gram are =)} \\
 \therefore \angle DCE &= 90^\circ \text{ (} \angle\text{'s at a point} = 360^\circ\text{)} \\
 \therefore x &= \frac{180^\circ - 90^\circ}{2} \text{ (} \angle\text{'s opp = sides in isos } \triangle CDE\text{)} \\
 \therefore x &= 45^\circ \text{ (} \angle\text{sum } \triangle CDE\text{)}
 \end{aligned}$$

$$\text{ii) } \sin 45^\circ = \frac{3}{DE}$$

$$\therefore DE = \frac{3}{\frac{1}{\sqrt{2}}} = 3\sqrt{2}$$

$$\text{e) } \therefore \frac{1}{x^2} - 9 = 0$$

$$\therefore \frac{1}{x^2} = 9 \rightarrow x^2 = \frac{1}{9}$$

$$\therefore x = \pm \frac{1}{3}$$

Question 9.

$$\begin{aligned}
 \text{a) } \frac{(1 - \cos \theta)(1 + \cos \theta)}{\tan^2 \theta} &= \sin^2 \theta \times \cot^2 \theta \\
 &= \sin^2 \theta \times \frac{\cos^2 \theta}{\sin^2 \theta} \\
 &= \cos^2 \theta
 \end{aligned}$$

b) i) Use $D(0,5)$ and $E(2,1)$

$$m = \frac{1-5}{2-0} = -2$$

$$\therefore y - 5 = -2(x - 0)$$

$$y - 5 = -2x$$

$$\therefore 2x + y - 5 = 0$$

ii) $m = \tan 45^\circ = 1$

$\therefore A$ has coordinates $(0,2)$

$\therefore$ the y -intercept is 2

hence using $y = mx + b$ the equation is

$$y = x + 2$$

9b) iii) $2x + y - 5 = 0$

$$y = x + 2$$

$$2x + (x + 2) - 5 = 0 \rightarrow x = 1$$

$$\text{when } x = 1, y = 3 \rightarrow B(1, 3)$$

iv) Use Line K and put $y = 0$.

$$\therefore 2x - 5 = 0 \rightarrow x = \frac{5}{2}$$

$$\therefore C\left(\frac{5}{2}, 0\right)$$

v) Area of $OABC$ = area FBC - area

$$\text{Area of quad } OABC = \text{area } \triangle FBC - \text{area } \triangle OAF$$

$$= \frac{1}{2} \times 4.5 \times 3 - \frac{1}{2} \times 2 \times 2$$

$$= 4\frac{3}{4} \text{ u}^2$$

Question 10.

a) Number of sides (n) = $\frac{360^\circ}{24} = 15$

$$\therefore \text{interior angle sum} = (180 - 24) \times 15 = 2340^\circ$$

b) Gradients multiply to equal -1 .

$$\therefore -\frac{k}{3} \times \frac{5}{2} = -1$$

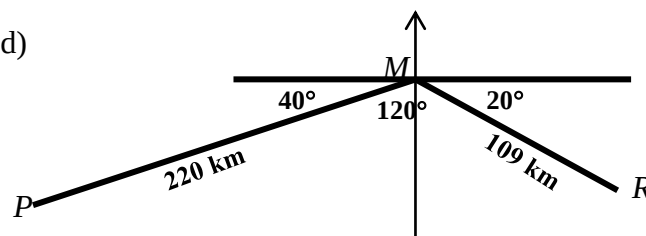
$$\therefore k = \frac{6}{5}$$

c) $\frac{9}{4x} = \frac{3x}{16}$ (transversals cut $\parallel$ lines in the same ratio)

$$\therefore 12x^2 = 144 \rightarrow x^2 = 12$$

$$\therefore x = \sqrt{12} \text{ (as } x > 0)$$

d)

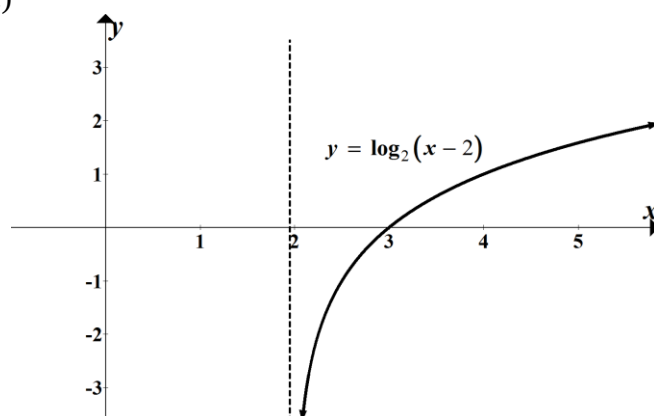


i) $\angle PMR = 230^\circ - 110^\circ = 120^\circ$

ii) $PR^2 = 220^2 + 109^2 - 2 \times 220 \times 109 \times \cos 120^\circ$

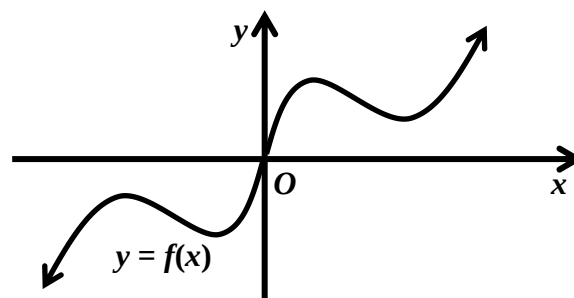
$$\therefore PR = 290 \text{ km (nearest km)}$$

e)



Question 11.

a)



b) When $x = -2$, $y = 4$

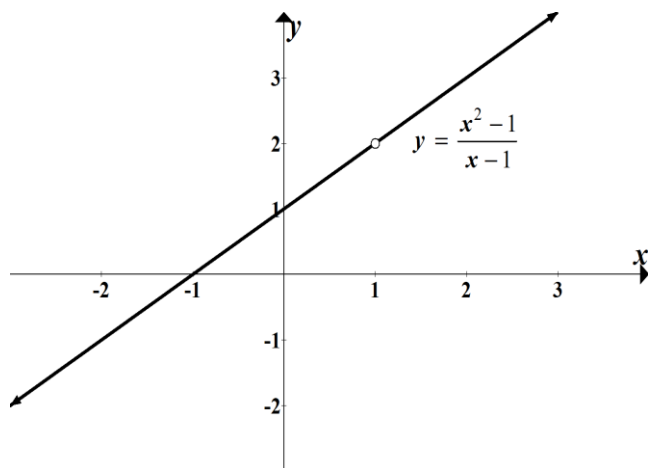
$$\therefore \text{width} = 4$$

$$\text{When } y = 4, x = 8$$

$$\therefore \text{length} = 8 + 2 = 10$$

$$\text{hence the area} = 40 \text{ u}^2$$

c)



d) $x^2 + y^2 - 6y + (-3)^2 = 7 + (-3)^2$

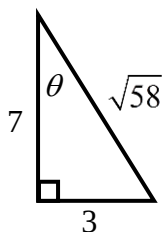
$$x^2 + (y - 3)^2 = 16$$

$\therefore$ centre is $(0, 3)$; radius = 4

e) $\tan \theta > 0 \rightarrow \theta$ is in the 1st or 3rd quad

$\sin \theta < 0 \rightarrow \theta$ is in the 3rd or 4th quad

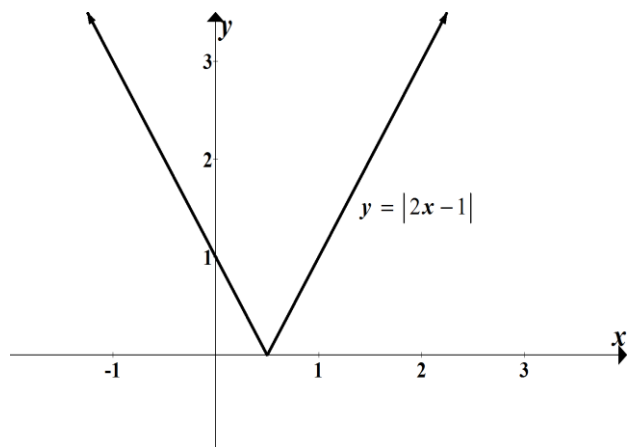
$\therefore \theta$ is in the 3rd quadrant and hence $\cos \theta < 0$.



$$\therefore \cos \theta = -\frac{7}{\sqrt{58}}$$

Question 12.

a)



b) $\angle BCA = 180^\circ - (a + b)$; ($\angle$ sum $\triangle ABC$)

$$\angle YZC = 180^\circ - c$$
; ($\angle$ sum straight line)

$$\therefore 180^\circ - (a + b) = x + (180^\circ - c)$$
; (exterior $\angle \triangle YZC$)

$$\therefore x = c - a - b$$

c) $16x^3 - 250 = 2(8x^3 - 125)$
 $= 2((2x)^3 - 5^3)$
 $= 2(2x - 5)(4x^2 + 10x + 25)$

d) i) $\angle NLM = \angle NKL + \angle KNL$ (ext $\angle$ of $\triangle NKL$)

$$\therefore 30^\circ = 15^\circ + \angle KNL \rightarrow \angle KNL = 15^\circ$$

hence $\triangle KLN$ is isosceles $\rightarrow KL = LN$

(sides opposite equal angles)

ii) Using $\triangle NML$ with $NM = 1$:

$$\sin 30^\circ = \frac{1}{NL} \rightarrow NL = 2$$

$$\therefore \boxed{KL = 2}$$
 from (i)

Also $\tan 30^\circ = \frac{1}{LM} \rightarrow \boxed{LM = \sqrt{3}}$

Now using $\triangle KNM$ with $NM = 1$:

$$\tan 15^\circ = \frac{1}{KM} = \frac{1}{KL + LM}$$

$$\therefore \tan 15^\circ = \frac{1}{2 + \sqrt{3}}$$

$$= \frac{1}{2 + \sqrt{3}} \times \frac{2 - \sqrt{3}}{2 - \sqrt{3}}$$

$$= \frac{2 - \sqrt{3}}{4 - 3}$$

$$= 2 - \sqrt{3}$$